

THANUSRI AENUGULA

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Education

Texas A&M University, College Station

Master of Science in Data Science

Aug 2025 - May 2027

College Station, TX

Sri Sivasubramaniya Nadar College of Engineering

Bachelor of Engineering in Electrical and Electronics Engineering

Aug 2021 - May 2025

Chennai, India

Experience

AMETEK - AMOT Controls | AI & Analytics Associate | Houston, TX

Aug 2026 - Present

- Built an **evaluation and observability layer** combining LLM-as-a-Judge scoring, a golden dataset of **500+ QA pairs**, human-in-the-loop review, and **LangSmith** tracing across agent runs, achieving **~95% answer accuracy** on queries and reducing debugging time.
- Delivered a core AI feature within 4 weeks by building a RAG chatbot in **Microsoft Copilot Studio** over product knowledge, enabling customer service representatives to access support information, improving response consistency and reducing manual search effort.
- Partnering with Supply Chain and Sales teams to identify automation opportunities and develop AI workflows and data solutions, enhancing cross-functional business processes.

H-E-B | Decision Science Analyst Intern | San Antonio, TX

May 2026 - Aug 2026

- Designed a **Next-Basket Recommendation system** by analyzing **11M+ customer transactions** to predict products customers are most likely to repurchase on their next shopping trip.
- Engineered **55+** temporal, behavioral, and basket-level features to capture purchase timing, shopping patterns, and customer replenishment behavior for personalized recommendations.
- Implemented and evaluated a Learning-to-Rank recommendation pipeline using **LightGBM LambdaRank**, transforming historical purchase data into personalized product rankings and achieving **75% recommendation accuracy**.
- Presented the recommendation framework and key findings to Data Science leadership, demonstrating a scalable methodology to improve basket completeness, customer loyalty, and shopping experience.

Texas A&M University | Research Assistant, Advanced Vision and Learning Lab | College Station, TX

Jan 2026 - May 2026

- Developed computer vision pipelines for automated plant growth analysis, transforming segmented plant imagery into quantitative growth measurements using **OpenCV**, image processing, and skeletonization algorithms to analyze complex plant architecture.
- Extracted **5+ plant traits**, including leaf area, width, path length, and structural measurements, and collaborated with AgriLife researchers to translate growth and morphology analyses into actionable insights for experimental planning and research decisions.

Daimler India Commercial Vehicles | Data Science Intern | Chennai, India

Jan 2025 - Jun 2025

- Collaborated with the Data-Driven Engineering team to develop machine learning solutions, streamlining **5+** vehicle testing workflows by integrating predictive analytics into existing testing pipelines.
- Constructed a **statistical twin** for Dynamic Brake Testing systems using historical data from **100+** vehicles, enabling predictive analysis and pre-screening of vehicle performance.
- Delivered a production-ready predictive solution that enabled testing teams to identify vehicle performance issues before physical testing, **reducing testing time by 80%**.
- Analyzed telematics and Dealer Management System data and developed **ETL pipelines** in **Azure Databricks** for real-time monitoring, driving pattern analysis, and fleet optimization.

Projects

Multi-Agent SQL Query Execution System [\[GitHub\]](#)

- Architected a **multi-agent LangGraph system** to automate natural-language-to-SQL analytics and reporting, with grounded reasoning, query validation, and constrained summarization, reducing manual effort by **60%** while achieving **100%** query execution validity.
- Deployed the multi-agent analytics system on **AWS EC2** with real-time inference and automated downloadable PDF report generation, reducing end-to-end time-to-insight from 2+ hours to under 2 minutes.

Predictive Maintenance of Wind Turbine Systems [\[GitHub\]](#)

- Benchmarked multiple models using sensor features and applied feature importance analysis to identify **34 key predictors**, including temperature, pitch angle, wind speed, voltage, and current-related features.
- Developed a hybrid **CNN-LSTM-GRU** model for Remaining Useful Life (RUL) prediction using the selected features, achieving an **R² of 0.91**, improving turbine failure forecasting for predictive maintenance planning.
- Designed a **Power BI** dashboard to monitor **10+ turbine health KPIs** and sensor metrics, using **DAX calculations** for key indicators.

Corrective RAG System for AI Knowledge Retrieval [\[GitHub\]](#)

- Developed a **Corrective RAG** pipeline using **LangGraph**, **FAISS**, and **OpenAI**, integrating semantic retrieval, document relevance grading, and conditional routing to validate retrieved context and improve factual grounding.
- Implemented query transformation and web search fallback to correct insufficient retrieval and reduce unsupported LLM responses, deploying the end-to-end workflow as a **Streamlit application** on **AWS EC2** for real-time inference.

Technical Skills

Tools:

Python, SQL, R, AWS, Microsoft Excel, Power BI, Tableau, PySpark, Databricks, Streamlit

Generative AI:

LangChain, LlamaIndex, LangGraph, Hugging Face Transformers, LLM Evaluation

Libraries & Frameworks:

Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, TensorFlow, Git

Techniques:

Time Series Analysis, Anomaly Detection, Clustering, Classification, Regression, Ensemble Methods (XGBoost, Random Forest), NLP, Computer Vision